Nikhil Dekate

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Skills

- **Programming Languages**: C, C++, Python, Core Java, SQL, HTML, XML
- **Software Tools:** GitHub, Jenkins, Visual Studio Code, SVN, ROBOT, Coverity, AndroidStudio.
- **Technical Skills:** Docker, Linux Internals, Shell Scripting, Socket programming, multi-threading, Device driver, prompt engineering, AI/ML (Basics).

WORK EXPERIENCE

Nokia Networks, Bangalore – Software Engineer (R&D 3) **Jan 2021 - Present**

- Developed a snapshot upload subsystem comprising two services that operate as a non-root user with limited privileges for file access and transfer. Created a plugin framework to manage input requests along with server configuration from the Network Management System (NMS). Implemented file transfers to the LSS server using the Paramiko library.
- Created a Diagnostic Check Framework that handles requests from the Network Management System (NMS) and provides diagnostic check results back to the NMS. This framework is part of the ne3s-agent, whichreceives managed object details from the NMS, performs basic validations, and passes the information to the diagnostic check plugin through the fpcmnms framework.
- Wrote entity specifications (EFS) and performed static code fixes using Coverity. Developed unit tests (UTs) utilizing the Google Test (gtest) framework and implemented functional tests (FT) using the ROBOT framework.
- As part of the fault management system, migrated startup scripts for Docker containers from Kubernetes to Swarm orchestration. Configured the containers to run as non-root users with limited privileges to enhance security and minimize potential risks associated with running processes as root.

KPIT, Pune – Software Engineer **July 2019 - November 2020**

- Implemented functionalities for the diagnostic framework of an all-wheel drive system. The project focused on controlling all four wheels and transmitting the required torque while ensuring the integration of safety features through the use of Electronic Control Units (ECUs). One ECU is responsible for delivering power to the rear unit, while another manages the clutch system. Communication between these components primarily utilizes the FlexRay bus, with CAN bus employed for driver-specific communication.

Appalto Electronics Pvt Ltd., Indore – Embedded Software Developer(R&D) **March 2018 - June 2019**

- Developed a driver for a Digital-to-Analog Converter (DAC) to accurately convert digital audio in WAV format to analog sound. The driver receives digital audio from a microcontroller, which is then transmitted to the Audio DAC via SPI. The DAC produces an analog output that is amplified and sent to the speaker, enabling sound playback in unmanned systems.
- Ported the entire weighing indicator code to a multitasking environment and enhanced the date-wise searching algorithm to deliver faster results.
- Implemented data management features for unmanned systems, improving overall data handling and accessibility.

ACADEMIC PROJECTS

- Trained multiple predictive models, including Random Forest and LightGBM, using datasets from various buildings. The datasets included features such as floor area, number of stories, and foundation type. The goal was to develop models capable of accurately predicting the electrical energy consumption of buildings based on these input parameters.
- Developed a Tic-Tac-Toe game on the Linux platform using C programming language and the GTK library for the user interface design. The game allows two players to compete against each other, with each player taking turns placing their mark (either X or O) on a 3x3 grid. The game automatically detects a winner or a tie condition and displays the result accordingly.
- Created a Linux block device driver that uses a 512 KiB RAM region as a virtual disk, partitions it into three primary partitions with the second further divided into three logical partitions, formats one primary partition as ext2 file system, mounts it, and enables reading/writing to the RAM-based file system through the custom driver, demonstrating the ability to create and manage a file system on a RAM disk using a Linux block device driver.

EDUCATION

PG Diploma in Embedded Systems Design, Pune --- CDAC July 2017 – Feb 2018

Bachelor of Engineering in Electrical, NAGPUR --- RCOEM July 2012 – July 2016